FINAL EXPLANATION: PHYSICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: PHYSICS | GRADE 10 | SUB-STRAND 1.5: MOMENTS OF EQUILIBRIUM	
Student Assessment Document	
Student Name	____
Class	____
Date	____

INSTRUCTIONS FOR STUDENTS

This Final Explanation document is your opportunity to show everything you have learned across the six lessons in this unit. Your goal is to answer the driving question fully and precisely, using the physics vocabulary, concepts, and evidence you have gathered. Read each section prompt carefully, then write your response in the space provided. Use the exemplar responses as a guide for the level of detail and scientific reasoning expected. A strong Final Explanation connects every idea back to the anchoring phenomenon — the Toppling Jiko Stool — and uses real Kenyan examples to support your thinking. Write in full sentences, include diagrams where prompted, and show all calculations clearly.

Section 1: The Anchoring Phenomenon — What Is Really Happening?	
Revisit the Toppling Jiko Stool phenomenon. In your own words, explain why the three-legged stool toppled when the 20-litre jerrycan was placed on one corner and nudged, but the wider, lower four-legged stool with the same jerrycan barely tilted and returned to upright. In your explanation, you must: (a) identify the turning force (moment) responsible for toppling, (b) explain the role of the base of support, and (c) explain the role of the centre of gravity. You do not need to use numbers here — focus on the physics concepts.	When the 20-litre jerrycan (roughly 20 kg, weight ≈ 200 N) was placed on the corner of the tall three-legged stool and a nudge was applied, the combined centre of gravity of the stool-plus-jerrycan system shifted until it moved outside the triangular base of support formed by the three legs. Once the centre of gravity was outside the base, the weight of the system created a net turning moment — a clockwise moment about the tipping edge (the leg nearest the jerrycan) — with no restoring anticlockwise moment to balance it. The stool therefore toppled. The wider, lower four-legged stool had a larger rectangular base of support. Even with the same jerrycan on one corner, the combined centre of gravity remained inside the wider base when nudged, so the weight produced a restoring moment — an anticlockwise moment about the tipping edge that pulled the stool back to upright. The key difference was not the force (the weight was identical) but the geometry: the width of the base and the height of the centre of gravity both determined whether the system was in stable or unstable equilibrium.

Section 2: Moments of Force — Calculating Turning Effects	
A market trader in Gikomba Market, Nairobi, uses a uniform wooden plank (length 2.4 m, weight 30 N) balanced on a pivot at its midpoint as a makeshift display shelf. She places a 5 kg bunch of bananas 0.8 m to the left of the pivot and a bag of maize flour of unknown mass 0.6 m to the right of the pivot. (a) Calculate the moment of the bananas about the pivot. (b) Using the principle of moments, calculate the mass of the maize flour bag needed to keep the plank in equilibrium. (c) State the principle of moments clearly in your own words. Show all working and units.	(a) Weight of bananas = 5 kg × 10 N/kg = 50 N. Moment of bananas (anticlockwise) = Force × perpendicular distance = 50 N × 0.8 m = 40 N·m anticlockwise. (b) Principle of moments states that for a body in equilibrium, the sum of clockwise moments about any pivot equals the sum of anticlockwise moments about the same pivot. Let the mass of the maize flour bag = m. Weight of maize flour bag = m × 10 N/kg. Clockwise moment = (m × 10) × 0.6. Setting clockwise moments equal to anticlockwise moments: (m × 10) × 0.6 = 40 N·m → 6m = 40 → m = 40 ÷ 6 ≈ 6.67 kg. The maize flour bag must have a mass of approximately 6.67 kg. (c) The principle of moments states that when an object is in rotational equilibrium, the total turning effect (moment) trying to rotate it clockwise about any point is exactly equal to the total turning effect trying to rotate it anticlockwise about the same point. This means all the turning forces are perfectly balanced — there is no net tendency to rotate.

Section 3: Centre of Gravity and Stability	
Explain the concept of centre of gravity and how its position determines whether an object is in stable, unstable, or neutral equilibrium. Use the Jiko stool and at least ONE other Kenyan example (such as a Maasai moran carrying a rungu, a bus on the Nairobi–Nakuru highway, or a cyclist balancing a load in Kisumu market) to illustrate each type of equilibrium. Include a clearly labelled diagram for each type of equilibrium showing the base of support, the centre of gravity, and the direction of the weight.	The centre of gravity (CG) of an object is the single point through which the entire weight of the object appears to act, regardless of the object's orientation. Its position depends on the distribution of mass — denser or heavier parts pull the CG towards them. Stable equilibrium: When a low, wide Jiko stool is tilted slightly, its CG rises and moves, but stays inside the base of support. The weight then produces a restoring moment that returns the stool to upright. A fully loaded matatu on the Nairobi–Nakuru highway with heavy luggage packed low in the boot is another example — its CG is low and well within the wide wheelbase, so it resists toppling on bends. [Diagram: wide base, CG shown low and inside the base, arrow for weight acting downward inside the base, restoring moment arrow shown.] Unstable equilibrium: The tall three-legged Jiko stool with the jerrycan on its corner is an example. When nudged, the CG moves outside the base of support and the weight creates an overturning moment — the stool falls further from equilibrium. A Maasai moran balancing his rungu vertically on one finger tip is in unstable equilibrium — the tiniest movement shifts the CG outside its tiny base and it falls. [Diagram: narrow base, CG shown high, arrow for weight acting outside the base, overturning moment arrow shown.] Neutral equilibrium: A sufuria (cooking pot) lying on its curved side on a flat floor is in neutral equilibrium. When rolled, the CG neither rises nor falls — it stays at the same height — so there is no restoring or overturning moment. The pot stays wherever it is placed. [Diagram: curved base, CG at constant height, no net moment arrow.]

Section 4: Conditions for Equilibrium

State and explain BOTH conditions necessary for an object to be in complete (static) equilibrium. Then apply both conditions to a real engineering or everyday Kenyan context: a sign board hanging outside a shop on Moi Avenue, Mombasa, supported by two vertical ropes attached to a horizontal beam. The sign has a weight of 120 N and is hung such that one rope is 0.3 m from the left end and the other is 0.3 m from the right end of a 1.2 m uniform beam (beam weight = 40 N). Calculate the tension in each rope. Show all working.	The two conditions for static equilibrium are: (1) Translational equilibrium — the sum of all forces acting on the object in every direction must equal zero ($\Sigma F = 0$). This means there is no net force causing the object to accelerate in any direction. (2) Rotational equilibrium — the sum of all moments (turning effects) about any point must equal zero ($\Sigma \tau = 0$). This means there is no net turning effect causing the object to rotate. Applying both conditions to the signboard: Let tension in left rope = T_1 and tension in right rope = T_2. Total downward force = weight of beam + weight of sign = 40 N + 120 N = 160 N. The sign is hung at the centre (0.6 m from each end). Condition 1 ($\Sigma F = 0$, vertical): $T_1 + T_2 = 160$ N. Condition 2 ($\Sigma \tau = 0$, taking moments about the left rope attachment point at 0.3 m from left end): Clockwise moments: Weight of beam acts at centre of beam = 0.6 m from left end, so distance from left rope = 0.6 − 0.3 = 0.3 m → moment = 40 × 0.3 = 12 N·m. Weight of sign acts at centre = distance from left rope = 0.3 m → moment = 120 × 0.3 = 36 N·m. Total clockwise moment = 12 + 36 = 48 N·m. Anticlockwise moment: T_2 acts at right rope, 0.6 m from left rope → moment = $T_2 \times 0.6$. Setting equal: $T_2 \times 0.6 = 48$ → $T_2 = 80$ N. From Condition 1: $T_1 = 160 - 80 = 80$ N. Both ropes carry equal tension of 80 N because the sign is centred and the beam is uniform — confirming the symmetry of the setup.

Section 5: Answering the Driving Question

Now answer the driving question directly and completely: 'Why do some objects stay balanced while others topple — and how can we use the physics of turning forces to predict and control whether an object will be stable or fall?' Your answer must: (a) connect the physics of moments, centre of gravity, and conditions for equilibrium into one coherent explanation; (b) explain at least TWO design strategies engineers or everyday Kenyans use to improve stability; and (c) reflect on what surprised you or changed your thinking during this unit.	Whether an object stays balanced or topples is determined by the interplay of three connected ideas: turning forces (moments), the location of the centre of gravity, and the conditions for equilibrium. A moment is the turning effect of a force about a pivot, calculated as $M = F \times d$. For an object to remain balanced, two conditions must be met simultaneously — the net force on it must be zero AND the net moment about any point must be zero. The centre of gravity is crucial because it is the point through which the object's entire weight acts. If the vertical line through the centre of gravity falls within the base of support, the weight produces a restoring moment when the object is tilted — it returns to upright (stable equilibrium). If it falls outside the base, the weight produces an overturning moment and the object topples. This is precisely what happened with the Jiko stool: the tall, narrow stool's combined centre of gravity (stool plus jerrycan) was pushed outside its small triangular base of support by the nudge, creating an unbalanced clockwise moment that caused it to topple. The wider, lower stool's centre of gravity stayed inside its larger rectangular base, so a restoring moment brought it back upright. We can use this physics to predict and control stability through deliberate design: (1) Lowering the centre of gravity — engineers design Kenyan matatus and buses with heavy engines and chassis low to the ground and warn against overloading roof racks, because a lower CG means greater stability on the winding roads of the Rift Valley escarpment. (2) Widening the base of support — a three-legged Jiko stool is less stable than a four-legged stool with legs spread wide; similarly, a Maasai moran spreads his feet apart when carrying a heavy load to widen his base. Both strategies increase the range of tilt before the CG escapes the base, making toppling far less likely. What surprised me most in this unit was realising that toppling has nothing to do with how heavy the object is — a light but top-heavy object topples far more easily than a heavy but low object. It is all about geometry and the position of the centre of gravity relative to the base of support.

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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Conceptual Understanding of Moments & Equilibrium	Accurately defines and applies moments ($M = F \times d$), the principle of moments, and both conditions for static equilibrium. All key terms (moment, pivot, equilibrium, centre of gravity, base of support) are used correctly and consistently throughout. Explains the distinction between stable, unstable, and neutral equilibrium with precision.	Correctly defines and applies most concepts. Minor errors in terminology or one condition for equilibrium is incompletely stated. Distinguishes between types of equilibrium but with limited depth.	Shows partial understanding of moments or equilibrium. Key terms are confused or missing. Only one condition for equilibrium is mentioned. Types of equilibrium are listed without meaningful explanation.
Mathematical Problem-Solving (Calculations)	All calculations in Sections 2 and 4 are fully correct with clear working, correct units, and appropriate significant figures. The principle of moments is explicitly applied as a reasoning step before calculating. Results are interpreted in context (e.g., what the answer means for the trader or the signboard).	Most calculations are correct. Working is shown but may skip one reasoning step. Units are mostly correct. Minor arithmetic errors do not obscure the method. Results are stated but not always interpreted in context.	Attempts calculations but makes significant errors in applying $M = F \times d$ or the principle of moments. Working is incomplete or unclear. Units are missing or incorrect. No contextual interpretation of results.
Connection to the Anchoring Phenomenon	The Toppling Jiko Stool phenomenon is explicitly and accurately referenced in every section. The student uses the phenomenon to anchor each new concept (moments, CG, stability, equilibrium conditions) and clearly explains the physical reason for the difference between the two stools using correct physics vocabulary.	The phenomenon is referenced in most sections and used to illustrate key ideas. The difference between the two stools is explained correctly but with limited use of precise vocabulary (e.g., 'overturning moment' or 'restoring moment' may be absent).	The phenomenon is mentioned in the introduction but not consistently connected to physics concepts across sections. The explanation of why one stool topples and the other does not is incomplete or relies on everyday language rather than physics.
Use of Diagrams and Visual Representations	Diagrams for all three types of equilibrium (Section 3) are present, clearly labelled, and scientifically accurate. Each diagram shows: the base of support, the centre of gravity, the line of action of weight, and the direction of the net moment (restoring or overturning). Diagrams are neat and support the written explanation.	Diagrams are present for at least two types of equilibrium and are mostly accurate. Most required labels are included. Diagrams are legible and generally support the explanation, though one or two labels may be missing.	Only one diagram is attempted or diagrams are present but largely unlabelled. The base of support, CG, or weight direction is missing from most diagrams. Diagrams do not clearly support the written explanation.
Real-World Application and Kenyan Context	At least three distinct Kenyan contexts (beyond the Jiko stool) are used accurately and relevantly across the Final Explanation — e.g., market traders, matatus, buses, cyclists, Maasai morans, or local buildings. Design strategies in Section 5 are clearly connected to real	At least two Kenyan contexts are used relevantly. Design strategies are identified and mostly explained using physics, but the connection to Kenyan practice is surface-level in one instance.	Only one Kenyan context is used, or contexts are mentioned but not connected to the physics concepts being discussed. Design strategies are listed without physics explanation.

	Kenyan engineering or everyday practice and are explained using correct physics principles.		